Question

Given $\mathrm{pK_a(HF)} = 3.17$; the equilibrium constant for the reaction $\mathrm{HF + F^- \rightleftharpoons HF_2^-}$ is $\mathrm{K_c^\circ} = 3.98$.

(1) To $100\ \mathrm{mL}\ 0.200\ \mathrm{mol \cdot L^{-1}\ HF}$, solid KOH is added (volume change is negligible, same below) to obtain buffer solution 1 with $\mathrm{pH} = 3.50$. The mass of KOH required is u g.

(2) Given $\mathrm{K_{sp}(LiF)} = 1.84 \times 10^{-3}$, assuming the pH of the solution remains unchanged, and using the mass balance relationship of the solution, the solubility s $(\mathrm{g \cdot L^{-1}})$ of LiF in solution 1 can be calculated as $v\ \mathrm{g \cdot L^{-1}}$.

(3) Assuming that solid NaF is added to buffer solution 1 instead, such that $[\mathrm{Na^+}]$ is equal to the $[\mathrm{Li^+}]$ calculated in the previous question, then the pH at equilibrium is w.

Then vw/u is approximately (If the calculated result has an error greater than 5% from all options, choose A)

- **A.** All other options are incorrect
- **B.** 1.47
- **C.** 2.08
- **D.** 0.26
- **E.** 1.92
- **F.** 4.06
- **G.** 3.88

H. 0.67

I. 3.12

J. 2.50

K. 4.44

L. 5.61

M. 6.06

N. 1.71

O. 3.54

P. 7.89

Answer

Correct Answer: C

Detailed Explanation

(1)

Let the final $[\mathrm{F}^-] = \mathrm{x\ mol \cdot L^{-1}}$

$[\mathrm{HF}] = (10^{-3.50}/10^{-3.17})\mathrm{x} = 0.468\mathrm{x\ mol \cdot L^{-1}}$

$[\mathrm{HF_2^-}] = 3.98 \cdot 0.468\mathrm{x} \cdot \mathrm{x} = 1.86\mathrm{x}^2\ \mathrm{mol \cdot L^{-1}}$

Material balance: $[\mathrm{F}^-] + [\mathrm{HF}] + 2[\mathrm{HF_2^-}] = (1 + 0.468)\mathrm{x} + 2 \cdot 1.86\mathrm{x}^2 = 0.200\ (\mathrm{mol \cdot L^{-1}})$

Solving for $x = 0.107$, $[\mathrm{F}^-] = 0.107\ \mathrm{mol \cdot L^{-1}}$, $[\mathrm{HF}] = 0.050\ \mathrm{mol \cdot L^{-1}}$, $[\mathrm{HF_2^-}] = 0.021\ \mathrm{mol \cdot L^{-1}}$

CHECKPOINT 2 PTS

$[\mathrm{F}^-] = 0.107\ \mathrm{mol \cdot L^{-1}}$, $[\mathrm{HF_2^-}] = 0.021\ \mathrm{mol \cdot L^{-1}}$

Ignoring $[\mathrm{OH}^-]$, $[\mathrm{K}^+] \approx [\mathrm{F}^-] + [\mathrm{HF_2^-}] - [\mathrm{H}^+] = 0.128\ \mathrm{mol \cdot L^{-1}}$, $\mathrm{m(KOH)} = 0.72\ \mathrm{g}$

CHECKPOINT 1 PTS

$\mathrm{m(KOH)} = 0.72\ \mathrm{g}$

(2)

If pH remains constant, the relationship between F, HF, and $\mathrm{HF_2}$ derived from the previous question remains unchanged.

Let $[\mathrm{F}] = \mathrm{y\ mol \cdot L^{-1}}$, $[\mathrm{Li^+}] = (1.84 \times 10^{-3}/\mathrm{y})\ \mathrm{mol \cdot L^{-1}}$

Material balance: $[\mathrm{F}] + [\mathrm{HF}] + 2[\mathrm{HF_2}] = \mathrm{c(HF)} + [\mathrm{Li^+}]$

CHECKPOINT 1 PTS

Material balance: $[\mathrm{F}] + [\mathrm{HF}] + 2[\mathrm{HF_2}] = \mathrm{c(HF)} + [\mathrm{Li^+}]$

$(1 + 0.468)y + 2 \cdot 1.86y^2 = 0.200 + 1.84 \times 10^{-3}/y$

Solving for $y = 0.114$, $[\mathrm{Li^+}] = 0.0161\ \mathrm{mol \cdot L^{-1}}$, $\mathrm{s} = 0.42\ \mathrm{g \cdot L^{-1}}$

CHECKPOINT 2 PTS

$\mathrm{s} = 0.42\ \mathrm{g \cdot L^{-1}}$

(3)

At this point, $[\mathrm{Na^+}] = 0.0161\ \mathrm{mol \cdot L^{-1}}$, and the analytical concentration of fluorine is $0.200 + 0.0161 = 0.216\ (\mathrm{mol \cdot L^{-1}})$

At this point, the solution is equivalent to $0.144\ \mathrm{mol \cdot L^{-1}}\ \mathrm{MF(KF + NaF)} + 0.072\ \mathrm{mol \cdot L^{-1}}\ \mathrm{HF}$

Let $[\mathrm{HF}] = \mathrm{a\ mol \cdot L^{-1}}$, $[\mathrm{F^-}] = \mathrm{b\ mol \cdot L^{-1}}$, $[\mathrm{HF_2^-}] = 3.98\mathrm{ab\ mol \cdot L^{-1}}$, $[\mathrm{H^+}] = 10^{-3.17}(\mathrm{a/b})\ \mathrm{mol \cdot L^{-1}}$

Material balance: $[\mathrm{HF}] + [\mathrm{F^-}] + 2[\mathrm{HF_2^-}] = \mathrm{a + b} + 2 \cdot 3.98\mathrm{ab} = 0.216\ (\mathrm{mol \cdot L^{-1}})$ (*)

CHECKPOINT 1 PTS

Material balance: $[\mathrm{HF}]+[\mathrm{F}^-]+2[\mathrm{HF}_2^-]=\mathrm{a}+\mathrm{b}+2\cdot 3.98\mathrm{ab}=0.216\ (\mathrm{mol}\cdot\mathrm{L}^{-1})$

Proton balance: $[\mathrm{HF}]+[\mathrm{HF}_2^-]+[\mathrm{H}^+]=\mathrm{a}+3.98\mathrm{ab}+10^{-3.17}(\mathrm{a}/\mathrm{b})=0.072\ (\mathrm{mol}\cdot\mathrm{L}^{-1})$ (**)

CHECKPOINT 1 PTS

Proton balance: $[\mathrm{HF}]+[\mathrm{HF}_2^-]+[\mathrm{H}^+]=\mathrm{a}+3.98\mathrm{ab}+10^{-3.17}(\mathrm{a}/\mathrm{b})=0.072\ (\mathrm{mol}\cdot\mathrm{L}^{-1})$

From (*), $\mathrm{a}=0.072/(1+3.98\mathrm{b}+10^{-3.17}/\mathrm{b})$

Substituting into (**): $0.072(1+7.96\mathrm{b})/(1+3.98\mathrm{b}+10^{-3.17}/\mathrm{b})+\mathrm{b}=0.216$

Solving for $\mathrm{b}=0.121$, $\mathrm{a}=0.0484$

$\mathrm{pH}=3.17-\lg(\mathrm{a}/\mathrm{b})=3.57$

CHECKPOINT 2 PTS

$\mathrm{pH}=3.57$

$u=0.72, v=0.42, w=3.57$, therefore $vw/u=2.08$, choose C